Binary Codes & Arithmetic

1. Classification of Binary Codes

Theory:

Binary codes are broadly categorized into alphanumeric codes (for text and characters) and numeric codes (for mathematical operations and number representation).

- **Alphanumeric Codes:**
  - **ASCII:** 7-bit code (128 values). Represents 26 upper-case, 26 lower-case, and special characters.
  - **EBCDIC:** 8-bit code ($2^8 = 256$ values).
  - **Unicode:** Available in 8-bit, 16-bit, and 32-bit formats to represent global character sets.
- **Numeric Codes (BCD - Binary Coded Decimal):**
  - Represent decimal numbers directly in binary.
- **Weighted Codes:** Each bit position has a specific mathematical weight. **Examples:**
  - `8421` (Standard BCD),
  - `631-1`
  - `2421`
  - `3321`
- **Non-Weighted Codes:** Positional weights do not apply.
  - **Gray Code:** A unit distance code where only a single bit changes between two successive numbers.
  - **Excess-3 Code:** Created by adding 3 (`0011`) to the standard 8421 BCD value.

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2. Standard Conversions Table

Here is the reference table you mapped out for Decimal numbers 0 through 9:

Decimal	Binary	Gray Code	Excess-3 Code
0	0000	0000	0011
1	0001	0001	0100
2	0010	0011	0101
3	0011	0010	0110

Decimal	Binary	Gray Code	Excess-3 Code
4	0100	0110	0111
5	0101	0111	1000
6	0110	0101	1001
7	0111	0100	1010
8	1000	1100	1011
9	1001	1101	1100

3. Gray Code Conversions

Theory:

Conversions between Binary and Gray Code rely on **XOR Addition** (Exclusive OR). In XOR logic, if the bits are the same, the result is `0`. If the bits are different, the result is `1`.

0	0	0
0	1	1
1	0	1
1	1	0

Binary to Gray Code

Rule: Bring down the Most Significant Bit (MSB) directly. Then, XOR each consecutive pair of adjacent bits in the binary number to get the next Gray code bit.

(Convert Binary 5 to Gray Code):

$$\begin{array}{rccccccc} \text{Binary:} & 0 & \oplus & 1 & \oplus & 0 & \oplus & 1 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow \quad \text{(XOR adjacent bits)} \\ \text{Gray:} & 0 & & 1 & & 1 & & 1 \end{array}$$

Gray Code to Binary

Rule: Bring down the MSB directly. Then, XOR your *newly calculated* binary bit diagonally with the *next* Gray code bit to get the next binary bit.

(Convert Gray Code 6 back to Binary):

$$\begin{array}{rccccccc} \text{Gray:} & 0 & & 1 & & 0 & & 1 \\ & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow \quad \text{(Diagonal XOR)} \\ \text{Binary:} & 0 & & 1 & & 1 & & 0 \end{array}$$

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4. Excess-3 Code Conversion

Theory:

To convert a decimal number to Excess-3 code, you simply take its standard 4-bit Binary (BCD) representation and add `3` (which is `0011` in binary) to it.

(Convert Decimal 0 to Excess-3):

$$\begin{array}{rll} & 0000 & (\text{Decimal } 0) \\ + & 0011 & (\text{Add } 3) \\ \hline & 0011 & (\text{Excess-3 Code}) \end{array}$$

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5. BCD Addition

Theory:

In BCD (Binary Coded Decimal), each decimal digit is represented by exactly 4 bits. Because 4 bits can hold up to the number 15 (`1111`), but decimal digits only go up to 9 (`1001`), the binary values `1010` through `1111` (10 through 15) are considered **invalid**.

The BCD Addition Rule:

1. Add the numbers using standard binary addition.
2. If the 4-bit sum is ≤ 9 (`1001`) and there is no carry out, the result is **valid**.
3. If the 4-bit sum is > 9 OR if it generates a carry into the next nibble, the result is **invalid**. You must add `6` (`0110`) to that 4-bit group to correct it.

(Example 1: Valid Result, 2 + 6):

$$\begin{array}{rll} & 0010 & (2) \\ + & 0110 & (6) \\ \hline & 1000 & (8 \rightarrow \text{Valid}, \ \leq 9) \end{array}$$

(Example 2: Invalid Result, 3 + 8):

$$
\begin{array}{rrll}
 & & 0011 & (3) \\
+ & & 1000 & (8) \\
\hline
 & & 1011 & (11 \rightarrow \text{Invalid},\ > 9) \\
+ & & 0110 & (\text{Add 6 to correct}) \\
\hline
 & 0001 & 0001 & (1 \text{ and } 1 \text{ in BCD} = 11)
\end{array}
$$

(Example 3: Invalid Carry-Out Result, 8 + 8):

$$
\begin{array}{rrll}
 & & 1000 & (8) \\
+ & & 1000 & (8) \\
\hline
 & & 10000 & (16 \rightarrow \text{Invalid carry generated}) \\
+ & & 0110 & (\text{Add 6 to correct}) \\
\hline
 & 0001 & 0110 & (1 \text{ and } 6 \text{ in BCD} = 16)
\end{array}
$$